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Visions in the Machine: Automated Tagging of the William Blake Archive

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Abstract

This paper investigates how multimodal AI can support — not replace — scholarly description of William Blake's visual art. Using the William Blake Archive as a foundation, I develop a conceptual framework that distinguishes three levels of visual analysis (pattern recognition, motif identification, and iconographic interpretation) and use it to evaluate a working retrieval prototype built on Contrastive Language-Image Pre-training (CLIP) embeddings and Facebook AI Similarity Search (FAISS) indexing over a 200-work stratified corpus. The paper defends three claims. First, that AI-assisted tagging is best understood as retrieval support rather than automated interpretation: it surfaces candidate matches that scholars then validate. Second, that performance degrades predictably across the three tiers: strong at object-level detection, moderate at motif-level pattern recognition, weak at Blake-specific iconography, and that this gradient is itself a useful empirical finding for archives weighing adoption. Third, that the responsible integration of AI tags into an existing scholarly archive requires explicit provenance, confidence scoring, human-in-the-loop validation, and audit logging, and that these requirements can be operationalized concretely. The paper does not claim to automate iconographic interpretation, to provide ground-truth identifications, or to generalize beyond the 200-work pilot; results should be read as alignment with the Archive's descriptive practice on a stratified sample, not as validated interpretation. The contribution is a conceptual framework, a tested prototype, and an integration design that other archives confronting dense symbolic corpora can adapt.

1. Introduction: The Challenge of Blake's Visions and the Promise of AI

William Blake (1757-1827) remains a singular figure in the annals of art and literature, a poet, painter, and printmaker whose work defies easy categorization. Operating primarily outside the mainstream artistic conventions of his time, Blake forged a unique visual and textual language to explore profound themes of spirituality, mythology, politics, and the human psyche [Frye 1947]. His art is characterized by an idiosyncratic blend of influences, drawing on yet radically transforming biblical narratives, Miltonic epic, Neoclassical forms, and Gothic sensibilities [Damon 1965]. The result is a body of work marked by its visionary intensity and interpretive depth. Central to the challenge of engaging with Blake's visual art is its characteristic complexity. His compositions are often densely populated with figures, interwoven with text, and structured according to symbolic rather than naturalistic principles. Blake developed a personal mythology populated by figures like Urizen (representing reason and law), Los (imagination and time), and Orc (rebellion and energy), whose meanings can shift across different works and contexts [Damon 1965] [Billingsley 2018]. This intricate symbolism, combined with his distinctive linear style and unconventional use of space, demands careful and informed interpretation. 1

One prime example of this complexity is *The Vision of the Last Judgement*. Extant in several versions and known primarily through detailed descriptions, this monumental conception presents a swirling vortex of figures that embody Blake's intricate cosmology and eschatology [Billingsley 2018]. Its sheer density, symbolic richness, and variations across related works pose significant analytical challenges for traditional art-historical methods, making it a compelling 2

test case for the potential application of computational techniques [Spratt and Elgammal 2014]. The study of Blake has been revolutionized by the William Blake Archive (the Archive), a landmark digital humanities project launched in the mid-1990s [Eaves et al. 2019]. The Archive provides unprecedented access to high-resolution digital reproductions of Blake's vast and widely dispersed works, including illuminated books, paintings, drawings, manuscripts, and commercial engravings. Its primary goal has been to establish a scholarly electronic edition, reuniting works scattered across global collections and providing detailed bibliographic, historical, and descriptive information for each object [William Blake Archive 2024].

Beyond serving as a digital repository, the Archive represents a structured dataset of significant potential for computational analysis. It meticulously documents provenance, physical characteristics, and relationships between different states or versions of works, accompanied by extensive textual transcriptions and scholarly descriptions. This digital infrastructure, while primarily designed for scholarly access and textual scholarship, offers a foundation upon which new computational methods, including those from artificial intelligence (AI), might be built to explore Blake's visual art in novel ways [Fiorucci et al. 2020] [Impett and Offert 2023]. 3

It is crucial to distinguish among three levels of visual analysis to structure this investigation. At the most basic level, computer vision systems engage in pattern recognition, detecting edges, textures, shapes, and colors in pixel data. The Archive, however, operates at the motif level, recording that a particular visual pattern represents a "tree" or "figure" rather than documenting its specific visual characteristics. Beyond this lies symbolic interpretation, understanding that a tree might represent the Tree of Life or natural innocence in Blake's mythology. This paper focuses primarily on how AI can bridge the gap between pattern recognition and motif identification, while acknowledging that symbolic interpretation remains fundamentally a humanistic endeavor. 4

1.1 Scope and Contributions

This paper makes three distinct contributions at different levels of completion: 5

1. A conceptual framework for understanding how AI can bridge pattern recognition and motif identification within established archival practices, distinguishing what machines can reliably detect from what requires human judgment.
2. A functional retrieval prototype demonstrating sub-millisecond similarity search across a 200-work corpus representing Blake's artistic range, with quantitative evaluation across three tiers of tagging complexity.
3. An exploratory case study applying iterative analytical methods to *The Vision of the Last Judgement*, demonstrating how regionally-balanced, source-weighted taxonomies can enhance motif detection in densely symbolic compositions.

I would like to pause here and state what this paper does not claim. To set clear expectations for a topic that invites overstatement, I explicitly state the boundaries of these contributions. This study investigates retrieval support, not automated interpretation. The system suggests candidate tags and visually similar works; it does not determine which figure is Urizen, decide what a given composition's serpent symbolizes, or produce iconographic readings. Where I report quantitative performance, I read those numbers as alignment with the Archive's existing descriptive practice on a stratified 200-work sample, not as a ground-truth interpretation of Blake. The case study of *The Vision of the Last Judgement* demonstrates how iterative computational passes can support a scholar's analysis of a single dense composition; it does not validate the method across Blake's full oeuvre. Generalization beyond the pilot would require multi-annotator ground truth, external expert review, and corpus-wide scaling tests that are out of scope here. 6

I frame this work as a pilot study: a proof-of-concept that establishes feasibility for a clearly bounded retrieval task, rather than a production system for the full Archive. The 200-work corpus was stratified from 715 works available in the Archive to represent diversity across Blake's oeuvre while remaining tractable for intensive methodological development; the rationale for that sample size, and the kinds of works it cannot yet speak to, are discussed in Section 6.2 and revisited under Limitations. 7

1.2 Operational Definition of "Tagging"

Throughout this paper, “tagging” refers to the computational process of associating image regions or whole images with descriptive labels from a defined vocabulary. I distinguish three levels of tagging complexity: 8

1. Object-level tagging identifies concrete visual elements (“serpent,” “book,” “human figure”) without interpretive claims about their meaning.
2. Motif-level tagging identifies compositional patterns and conventional postures (“outstretched arms,” “ascending figure,” “vortex composition”) that recur across Blake’s work.
3. Iconographic tagging assigns Blake-specific identifications (“Urizen,” “Los,” “Last Judgement throne”) that require contextual knowledge of his mythology.

This hierarchy corresponds to increasing interpretive demands: object-level tagging lies closest to pattern recognition, while iconographic tagging approaches interpretation. My pilot implementation demonstrates feasibility across all three levels and confirms that performance degrades as tagging approaches the interpretation stage, with significant implications for human-AI collaboration in digital art history. 9

1.3 Research Questions

This paper addresses the following core research question: 10

How can current AI techniques assist scholars in the identification and cataloging of visual motifs within William Blake’s paintings by automating the translation from visual patterns to motif labels, and how can this pattern-to-motif mapping be integrated with the existing descriptive infrastructure of the William Blake Archive? 11

1.4 Implications for Scholarly Practice

If successfully developed and validated, AI-assisted tagging would open three currently impractical lines of inquiry. Scholars could trace the frequency and evolution of specific visual elements, such as the serpent motif, across Blake’s entire oeuvre at a scale manual cataloging cannot reach, revealing patterns invisible to inspection of single works. Similarity search could surface unexpected visual affinities, linking, say, a figure in the Job illustrations to a compositionally similar figure in the Dante series and prompting comparative analyses that would not otherwise be initiated. And for works existing in multiple states or copies, automated comparison could highlight systematic variations in Blake’s treatment of specific elements across versions. None of these capabilities replaces interpretive scholarship; they redirect it from exhaustive cataloging (which the system can assist) toward interpretive synthesis (which remains human work). 12

This shift raises new responsibilities for archives: how to present AI-generated tags alongside human descriptions, how to indicate confidence levels, and how to prevent algorithmic suggestions from foreclosing interpretive possibilities rather than opening them. This inquiry is inherently interdisciplinary, situated at the intersection of art history, digital humanities, and computer science [Spratt and Elgammal 2014]. It seeks to bridge the gap between the nuanced interpretive methods of art history and the pattern-recognition capabilities of AI, exploring both the potential synergies and the inherent limitations of applying computational tools to such complex and symbolically rich visual material. 13

2. The William Blake Archive: Structure, Data, and Current Practices

The William Blake Archive is built upon a robust technical architecture designed to manage and present a large, complex corpus of digital objects and associated metadata [Eaves et al. 2019]. While specific technical details evolve, its foundation involves standard digital library technologies and adherence to metadata standards common in digital humanities. Works are meticulously cataloged with unique identifiers, enabling precise referencing. Crucially, the Archive model accommodates the complex relationships inherent in Blake’s output, such as linking different copies of illuminated books (which Blake produced individually, leading to variations) or relating preliminary sketches to finished works. 14

Metadata standards play a key role. The Archive utilizes the Text Encoding Initiative (TEI) guidelines for encoding Blake’s texts, ensuring scholarly rigor and interoperability [William Blake Archive 2024]. While TEI is primarily text- 15

focused, the Archive also incorporates extensive metadata about the visual works themselves, covering aspects like medium, dimensions, support, date, and provenance. The Archive uses XML extensively, employing a primary Document Type Definition (DTD) known as the Blake Archive Description (BAD) for encoding texts, bibliographic commentary, and illustration descriptions, while TEI is used for other materials, such as bibliographies, where physical artifact description is not central [Eaves et al. 2019]. For instance, the multiple descriptions and potential lost versions of *The Vision of the Last Judgement* necessitate a structure capable of handling such complex interrelations. Understanding this underlying data model is crucial for assessing how visual information is represented and whether it can be readily extracted for AI purposes.

The Archive provides detailed scholarly descriptions for many of its objects. Examining the entry for a complex work like the Petworth House version of *The Vision of the Last Judgement* reveals the nature of current practices. Descriptions are typically narrative and interpretive, written by experts to guide understanding [William Blake Archive 2024]. They often identify key figures, explain the overall scene, discuss symbolism, and relate the work to Blake's writings or other compositions. Fields like "Description" or potentially "Objects Represented" contain this information, often in free-text prose. 16

The vocabulary used in these descriptions is scholarly and interpretive, not necessarily a controlled vocabulary designed for machine processing. While terms might be consistent within a single description or by a single cataloger, variations in terminology across the entire archive are likely. Consistency depends heavily on editorial oversight and the specific cataloging guidelines employed over the project's long history [Eaves et al. 2019]. Furthermore, the granularity of existing descriptions varies. They excel at providing a holistic interpretation of the scene but may not systematically list every single figure, object, or motif depicted. The focus is often on salient or symbolically important elements rather than exhaustive visual inventory. Annotations directly linking specific parts of the image to descriptive text are generally absent. 17

2.1 Strengths and Limitations as AI Training Data

The Archive offers significant strengths as a potential data source for AI-driven analysis of Blake's art. It provides access to a large corpus of high-resolution images, essential for training deep learning models [Eaves et al. 2019]. The images are accompanied by rich contextual metadata, including detailed provenance, dating, and relationships to other works, which could potentially inform AI analysis. Furthermore, the extensive textual descriptions and transcriptions associated with many images offer a valuable resource, particularly for multimodal AI approaches [Radford et al. 2021]. 18

However, the Archive also presents limitations when considered as a direct source of training data for standard computer vision tasks. The primary challenge lies in the nature of the visual annotations. Visual descriptions are typically narrative and interpretive rather than structured lists of labels suitable for supervised learning [Spratt and Elgammal 2014]. There may be inconsistencies in terminology and granularity across the archive. Crucially, the lack of fine-grained annotations, such as bounding boxes or segmentation masks, which specify the location and extent of objects within the image, poses a significant hurdle for training object detection or segmentation models [Ren et al. 2015]. 19

This assessment points towards a structural characteristic of the archive: Its origins and primary goals — centered on scholarly access, textual encoding (TEI usage), and comprehensive bibliographic and historical metadata — likely result in textual data being far more structured and consistent than visual metadata from an AI perspective. Visual description, while rich, follows humanistic interpretive conventions rather than the systematic tagging protocols ideal for machine learning [Fiorucci et al. 2020]. 20

2.2 The Archive's Editorial Stance

The William Blake Archive maintains a deliberately restrained approach to interpretation, focusing on descriptive documentation that aims for scholarly objectivity. As the Archive's editors note, illustration descriptions are "[a]lthough information cannot be completely separated from interpretation, our emphasis is strongly on information and hence on description" [William Blake Archive n.d.]. The Archive engages in limited interpretation, such as labeling a figure as 21

“Urizen,” but avoids deep symbolic analysis. This approach provides the ground for rigorous interpretation, much as a letterpress text provides the ground for literary interpretation. This philosophy has important implications for AI integration. The Archive records the representation of motifs, not patterns. When a human cataloger identifies a “tree” in Blake’s work, they are making an interpretive leap from a specific visual pattern to a conceptual category that encompasses infinite possible visual instantiations. No two trees in Blake’s art, or in nature, share the same pattern, yet humans readily group these diverse patterns under the single motif label “tree.” This many-to-one mapping from patterns to motifs represents a fundamental challenge for computational approaches.

3. Blake’s Iconographic Complexity

3.1 Blake’s Iconographic System

A vast body of art historical scholarship is dedicated to deciphering William Blake’s complex and deeply personal iconography. Blake constructed a unique mythological system populated by figures such as Urizen (representing reason and law), Los (imagination and time), Orc (rebellion and energy), Enitharmon (spiritual beauty and female counterpart to Los), and many others [Damon 1965]. These figures are not static allegories but dynamic entities whose significance can evolve throughout Blake’s career and vary depending on the specific context of a work [Frye 1947]. 22

Key symbols of the serpent, compasses, flames, lilies, and anvils all recur throughout his oeuvre, imbued with layers of meaning drawn from biblical, Miltonic, Neoplatonic, alchemical, and contemporary political sources, all refracted through Blake’s unique visionary lens [Serdechnaya 2018]. Blake’s relationship with existing traditions was one of radical transformation. He engaged deeply with the Bible and the works of John Milton but reinterpreted them according to his own spiritual and political vision, often inverting traditional meanings [Frye 1947]. Similarly, while employing elements of Neoclassical linearity and Gothic energy, his style ultimately served his internal vision rather than adhering strictly to contemporary aesthetic norms [Billingsley 2018]. This synthesis of diverse sources into a highly personal symbolic language creates significant interpretive challenges. Meanings are often polysemous (having multiple interpretations), ambiguous, and resistant to straightforward decoding. Understanding Blake’s symbolism requires familiarity not only with his visual works but also with his poetry and prophetic writings, where these concepts are further elaborated [Damon 1965]. 23

3.2 Recurring Visual Motifs

Beyond specific symbols and named figures, Blake’s visual art is characterized by recurring visual motifs, postures, gestures, and compositional structures [Billingsley 2018]. Figures with dramatically outstretched arms, bodies bent in despair or constricted by bonds, dynamic vortex compositions suggesting spiritual energy or chaos, and layered planes creating a sense of shallow, stage-like space are common features. Blake’s distinctive style emphasizes strong contour lines, flattened perspective, and a powerful, often idealized or exaggerated, rendering of the human form [Frye 1947]. A key element of his visual strategy, particularly in the illuminated books, is the integration of text and image. Text does not merely caption the image, nor does the image simply illustrate the text; rather, they interact in complex ways, often weaving together on the page, with decorative elements blurring the lines between writing and drawing [Eaves et al. 2019]. These recurring visual elements and stylistic choices are not merely decorative; they are integral to the works’ meaning and affective power, conveying states of being, spiritual struggles, and cosmic events through a consistent, albeit evolving, visual vocabulary [Serdechnaya 2018]. 24

3.3 The Vision of the Last Judgement as Test Case

The Vision of the Last Judgement serves as an exemplary case study of Blake’s complexity [Billingsley 2018]. Known through several related works and detailed descriptions by Blake himself, the composition depicts a vast panorama of humanity rising to judgment or falling into damnation, presided over by Christ enthroned on the Mercy Seat, surrounded by apostles and heavenly figures. Below, the Book of Life is opened, Satan is wrapped in the Serpent’s coils, and figures representing various states of salvation and damnation populate the scene. The composition is dense, dynamic, and spatially complex, drawing heavily on traditional Last Judgement iconography but reinterpreted through Blake’s 25

personal mythology [Damon 1965].

Identifying specific figures requires knowledge of both biblical tradition and Blake's own symbolic system [Frye 1947]. Distinguishing between numerous nude or semi-nude figures based on subtle postural differences or their position within the overall symbolic program is challenging. Interpreting the relationships between figures — for example, the complex interplay between judgment, mercy, and the creative imagination — requires deep engagement with Blake's thought [Billingsley 2018]. Specific elements, like the precise identity of allegorical figures representing states of sin or redemption, or the exact meaning of certain gestures, remain subjects of scholarly debate [Serdechnaya 2018]. 26

Each of the demands set out above, from recognizing context-dependent symbols to distinguishing members of Blake's personal pantheon by subtle attributes, to parsing densely overlapping figures in compositions like *The Vision of the Last Judgement*, represents a different kind of computational challenge. Each of these concerns has been engaged to varying degrees by different strands of the technical literature. Computational iconography has begun to handle motif-level recognition for artists with shared conventional vocabularies but has largely avoided artists with idiosyncratic personal mythologies. Multimodal vision-language models have made flexible, prompt-based retrieval newly feasible for small-corpus archives. Similarity-retrieval infrastructure has matured to the point where latency and scale are no longer the constraints they once were. And the digital-humanities literature on AI evaluation has supplied the vocabulary needed to ask of any such system what it is doing, on whose behalf, and against what standard. The next section surveys these four conversations in turn, to locate precisely where the present paper enters them — and where it does not. 27

4. Related Work: AI, Computer Vision, and Cultural Heritage

Related work falls into four conversations that this paper joins: computational iconography and motif recognition in art history; multimodal vision-language models applied to cultural heritage; similarity retrieval over art image collections; and the digital-humanities literature on critical evaluation and bias in AI-assisted scholarship. Treating them together makes this paper's contribution easier to locate. The paper is the first, to my knowledge, to bring all four to bear on an artist with a dense personal mythology, and to do so under a framework that explicitly distinguishes what a computational system supports (retrieval) from what it does not (interpretation). 28

The four conversations are also a useful diagnostic for why Blake has remained outside the main computational-art-history literature to date. Each conversation handles part of what a Blake-scale tagging project would need; none has yet had to address them all at once. 29

4.1 Computational Iconography and Motif Recognition

The computational analysis of art-historical imagery has matured rapidly over the last decade. Foundational work by [Crowley and Zisserman 2014] established visual search over painting collections, demonstrating that off-the-shelf object detectors trained on natural photographs could transfer, with caveats, to artistic imagery. Subsequent work extended this into explicitly iconographic territory: [Milani and Fraternali 2020] released a dataset and convolutional model for iconography classification in paintings, and [Springstein et al. 2024] pushed the problem toward large-scale hierarchical classification, demonstrating that structured label spaces can substantially improve performance over flat classifiers. [Schneider and Vollmer 2024] provide an instructive model from an adjacent problem space: Their art-historical pose dataset required iterative refinement of annotation guidelines and quality-controlled multi-annotator review. Their work addresses pose estimation across a multi-collection dataset and does not engage with the Archive directly; I cite it here as a procedural template for how annotation guidelines could be developed if and when the Archive's curatorial team chose to extend descriptive metadata in this direction. 30

These projects share a common architecture. They start with a labeled corpus (often built with substantial editorial labor), train or fine-tune a deep model on that corpus, and report accuracy on held-out subsets. Their typical objects of study are stylistic or attributional [Johnson et al. 2008] or, in the iconographic case, drawn from broadly shared visual vocabularies such as saints, biblical scenes, and classical motifs whose iconographic conventions are stable across centuries and across many artists. The general-purpose computer-vision toolkit they draw on: convolutional 31

architectures such as ResNet [He et al. 2016], detection frameworks such as Faster R-CNN [Ren et al. 2015] and Mask R-CNN [He et al. 2017], and Vision Transformers [Dosovitskiy et al. 2020]. All of these techniques have been extended further by end-to-end detection models [Carion et al. 2020] and by general-purpose segmentation systems such as Segment Anything [Kirillov et al. 2023].

Three reasons explain why these methods have largely avoided artists with dense personal iconographies like Blake. First, the data requirement: A personal symbolic system requires not just images but extensive scholarly annotation that links visual features to specific meanings, and that annotation does not yet exist at the scale supervised methods need. Second, the evaluation problem: Standard accuracy metrics measure pattern-matching against a ground truth, but a meaningful Blake iconographic identification typically depends on context, on inter-textual evidence, and on contested scholarly judgment that no single ground truth captures [Spratt and Elgammal 2014]. Third, the interpretive problem: An artist whose imagery deliberately resists conventional iconography is precisely the artist for whom automated interpretation is least likely to succeed. The present paper does not attempt to remove these obstacles. It accepts them and proposes a framework in which AI supports retrieval and surfaces candidate matches that scholars validate, rather than producing iconographic readings directly. 32

4.2 Multimodal Vision-Language Models and Cultural Heritage

The most consequential recent shift in computer vision for cultural-heritage applications has been the emergence of multimodal vision-language models. CLIP (Contrastive Language-Image Pre-training) [Radford et al. 2021] and BLIP [Li et al. 2022] place images and natural-language text into a shared embedding space, allowing flexible retrieval and zero-shot classification without task-specific training. The practical effect is that art-historical applications no longer require a fully labeled dataset before useful experiments can begin: A researcher can write a tag prompt in plain English and retrieve candidate matches against an embedding index of an entire collection. 33

[Impett and Offert 2023] frame the broader claim that digital art history has matured beyond the digitization phase into a computationally distinct discipline, in part because multimodal models such as CLIP can reveal visual relationships across large corpora that resist manual analysis. The methodological turn this paper joins, then, has two features. Earlier supervised approaches surveyed by [Fiorucci et al. 2020] treated the labeled dataset as a prerequisite for analysis; the multimodal turn treats the prompt as the unit of investigation and the embedding as the substrate. This shift is particularly consequential for archives like the William Blake Archive, where the labor required to assemble a supervised dataset at scale has been the principal barrier to computational analysis. 34

To tailor a pre-trained model to a specific artist or lexicon, researchers continue to rely on adaptation methods suited for limited labeled data, including transfer learning [Dosovitskiy et al. 2020], few-shot learning [Wang et al. 2020], and weakly supervised techniques [Zhou 2018]. The implementation reported in this paper uses zero-shot CLIP with no fine-tuning, partly because the Archive does not yet provide labels in a form suitable for supervised training, and partly because zero-shot retrieval is precisely the operation the paper's contribution is designed to support. The trade-offs between zero-shot and fine-tuned configurations are discussed methodologically in Section 6.2. 35

4.3 Similarity Retrieval Over Art Image Collections

The third conversation this paper joins is the technical literature on similarity retrieval in art collections. [Seguin et al. 2016] demonstrated visual link retrieval across painting databases, showing that learned visual features could surface non-obvious compositional relationships among works in large heterogeneous corpora. [Garcia-Seco de Herrera et al. 2020] report the related ImageCLEF retrieval benchmarks that have set the practical standard for cross-domain image retrieval evaluation. On the indexing side, the FAISS library [Johnson et al. 2019] has made it routine to deploy sub-millisecond similarity search over millions of high-dimensional embeddings; for collections in the hundreds or low thousands of works typical of single-artist archives, retrieval latency has ceased to be a meaningful constraint. 36

More recent work on open-vocabulary detection [Minderer et al. 2022] closes part of the gap between similarity retrieval and tag-level identification, allowing systems to localize objects within an image based on natural-language prompts. The retrieval prototype in Section 6.4 of this paper draws on this lineage technically (CLIP embeddings indexed by 37

FAISS) and conceptually: The contribution is positioned as retrieval support, not as a novel detection architecture. What is new is the application to an artist whose iconographic vocabulary is unusually personal and unusually dense, and the explicit framing of retrieval support as the appropriate level of computational assistance for that case.

4.4 Critical Evaluation and Bias in AI-Assisted Scholarship

The fourth conversation is the digital-humanities literature on what AI does to scholarship when it is brought into it, and on what scholars should ask of computational tools before accepting their outputs. Crawford's (2021) account of the political economy of AI systems and Foka and colleagues' (2025) analysis of the bias loop in cultural-heritage ML pipelines both name the same problem from different angles: biases embedded in historical collection practices (what was preserved, described, and prioritized) are encoded into training data, amplified by model behavior, and then naturalized by deployment into archive interfaces [Crawford 2021] [Foka et al. 2025]. The Archive's own descriptive practice is not exempt from this loop, and any computational system trained on or evaluated against its descriptions inherits the priorities of its editorial history. Section 6.6 returns to this point as an evaluation-circularity disclosure. 38

On the evaluation side, [Powers 2011] provides the standard reference for precision, recall, F-measure, and the conditions under which each is appropriate; this paper relies on Powers's framing in Section 6.6 when reporting tier-stratified performance. [Spratt and Elgammal 2014] anticipated many of the cross-disciplinary tensions that still shape evaluation: Art historians and computer scientists tend to disagree about what a successful identification looks like, about how much interpretive content a tag should carry, and about what counts as a fair test case. The three-tier framework adopted in this paper (Section 1) is partly a response to Spratt and Elgammal's observation: By distinguishing object-level, motif-level, and iconographic-level tags, the framework makes the evaluation appropriate to the claim and makes the system's limits visible to readers from both disciplines. 39

4.5 What This Paper Adds

Across these four conversations, the present paper is positioned not as a technical demonstration but as a digital-humanities argument. It accepts that computational iconography has not yet engaged Blake, that the multimodal turn lowers the barrier to entry, that similarity retrieval is now technically routine at this scale, and that critical evaluation must remain central. Its contribution is to bring those four points together under a single framework: the three-tier distinction between pattern, motif, and iconographic interpretation, and to test that framework on a 200-work stratified Blake corpus. 40

Three things follow from that positioning. First, the paper does not claim to extend the state of the art in computer vision; the technical components (CLIP, FAISS, PCA visualization) are standard and intentionally so. Second, the paper does claim that applying those standard components to a corpus this iconographically demanding, under a framework that explicitly bounds what the system can do, yields findings that are useful to other archives confronting similar problems, most concretely, a measurable performance gradient across the three tiers and a concrete integration plan (Section 8) for AI-suggested tags alongside human descriptive scholarship. Third, the paper claims that the specifically humanities-side contribution, the framework, the integration plan, and the set of success criteria specified in Section 8, is the part most worth carrying forward to artists and corpora beyond Blake. 41

5. Data and Annotation in the Archive

Section 4 surveyed the technical literature on computer vision, multimodal models, similarity retrieval, and critical evaluation. The question now is operational: What does the William Blake Archive specifically provide, what does it not provide, and what follows for how a computational system can be built on top of it? This section is deliberately narrow. It does not restate general requirements for training computer vision models; those are already in the literature cited and would only duplicate Section 4. Instead, it maps the Archive's existing data against the analytical requirements established there, identifies the gaps that matter for this study, and sets up the methodological choices made in Section 6. 42

5.1 Mapping the Archive Against Analytical Requirements

The Archive's principal strength as a data source is the combination of image quality and scholarly description: High-resolution images of roughly 715 fully digitized works are paired with TEI-encoded textual descriptions written by Blake specialists over three decades of editorial work [Eaves et al. 2019] [William Blake Archive 2024]. For zero-shot retrieval against a pre-trained multimodal embedding model (as in Section 6), image volume and quality are not binding constraints. The binding constraints are in the label structure, not in the images. 43

Requirement	Archive Status	Gap Assessment
Image quantity	715+ works, 5,500+ images	Sufficient for transfer learning
Image quality	High-resolution digital surrogates	Excellent
Label type	Narrative descriptions	Requires extraction/structuring
Label consistency	Variable across catalogers	Moderate standardization needed
Spatial annotations	Generally absent	Significant gap
Text-image pairing	Rich descriptions available	Strong foundation for multimodal

Table 1. Blake Archive Data vs. AI Training Requirements: Mapping the William Blake Archive's existing data against the requirements of common computational analyses. The image-side of the Archive comfortably exceeds current model requirements; the operational gaps are concentrated on the label side.

5.2 The Label-Side Gap

Four interrelated gaps shape what is and is not possible at present. The Archive does not currently provide bounding boxes or segmentation masks of the kind that object-detection and segmentation models require. Scholarly descriptions use varied terminology for the same concept, such as “serpent,” “snake,” and “Satan-Serpent” which appear across descriptions as the cataloger's reading of a given figure shifts, and thus require substantial normalization before they can serve as supervised labels. Descriptions tend to characterize overall interpretation rather than exhaustively list every identifiable element, because that is what the Archive's editorial practice rightly prioritizes. And coverage is uneven across the corpus, with richly described works sitting alongside minimally annotated ones, reflecting the Archive's longstanding decision to prioritize editorial depth over uniform annotation. 44

The first two of these gaps are addressable by editorial labor and controlled vocabulary; they describe missing infrastructure, not missing scholarship. The second two are constitutive features of the Archive's mission and should not be “fixed” so much as worked with. Exhaustive labeling of every figure and object in every work was never the Archive's purpose, and a computational system that requires such labeling is asking the wrong thing of the Archive. A system that supports scholarly retrieval by surfacing candidate matches and leaving interpretation to the scholar does not need exhaustive supervised labels in the first place. 45

5.3 Implications for Method

Three methodological choices follow directly from the data state described above and govern the implementation in Section 6. First, the system is built on zero-shot multimodal embeddings rather than supervised classifiers, because the Archive's current labels do not support supervised training at scale. Second, evaluation is framed as alignment with the Archive's existing descriptive practice on a stratified sample, rather than as validation against an external ground truth; the Archive's descriptions are the most authoritative reference available, and any computational system trained on or evaluated against them inherits the priorities of its editorial history, a circularity disclosed in Section 6.6. Third, the three-tier framework introduced in Section 1 (object-level, motif-level, iconographic-level) is partly a response to the 46

granularity gap described in Table 1: By matching the evidentiary burden to the kind of tag being produced, the framework makes the system's performance interpretable even where the underlying labels are uneven.

6. Methods and Implementation

6.1 Methods at a Glance

For readers approaching this study from a humanities background, the technical vocabulary in this section can be reduced to five working terms. *CLIP* (Contrastive Language–Image Pre-training) is a neural model that places images and text into a shared vector space so that semantically related items lie near one another. An *embedding* is the coordinate of an item in that space. For this study, an embedding is a 512-dimensional vector summarizing what a model has learned to associate with a Blake image. *FAISS* is a software library that indexes embeddings to retrieve the nearest matches to a query in milliseconds. Other metrics include: *mAP* (mean Average Precision), which measures whether the most relevant tags are ranked near the top of a multi-label prediction and *NDCG* (Normalized Discounted Cumulative Gain) which measures whether the most relevant retrieved images are ranked highest. None of these terms should obscure the operation they describe: convert each image into a numerical fingerprint, store the fingerprints in a fast lookup structure, and rank candidate matches for a scholar to verify. 47

Throughout Section 6, I use the following scholar-in-the-loop scenario as a concrete anchor. A researcher studying the serpent motif submits a query, either an image of the lower-left register of *The Vision of the Last Judgement* or the text string “coiled serpent form.” The system computes the query’s embedding, retrieves the 20 nearest works in the corpus, returns them ranked by similarity with provenance metadata (AI-suggested vs. human-validated; confidence; source tier), and the researcher confirms or rejects each match. The same example is used in 6.3 to describe the index, in 6.4 to demonstrate iterative taxonomy expansion, and in 6.5 to interpret the evaluation metrics. 48

6.2 Methodological Framework

The methodological framework adopted in this paper uses pre-trained CLIP (ViT-B/32) in a zero-shot configuration, that is, without any fine-tuning on Blake-specific data. All similarity search, retrieval, and tag-candidate generation rely on CLIP’s original image–text embedding space. The discussion below of transfer learning, fine-tuning, few-shot learning, and supervised detection architectures (Faster R-CNN, Mask R-CNN, ViT) is included to situate the chosen approach within the broader space of options and to clarify what trade-offs would attend a future fine-tuned or supervised variant; it does not describe the system evaluated in Sections 6.4–6.6. Zero-shot CLIP was chosen because it permits flexible querying with natural-language tag prompts derived from Blake scholarship without requiring a labeled training set that the Archive does not currently provide. 49

A primary methodological approach involves leveraging transfer learning with powerful vision models pre-trained on large-scale image datasets [Dosovitskiy et al. 2020]. Models such as Vision Transformers (ViTs) or robust CNN architectures (e.g., ResNet families) serve as the starting point. The strategy involves fine-tuning these models on a curated dataset derived from the Archive [He et al. 2016]. 50

Given the symbolic nature of Blake’s art, multimodal models like CLIP offer a particularly promising avenue [Radford et al. 2021]: 51

- **Zero-Shot Tagging/Detection:** Using a pre-trained CLIP model directly, without fine-tuning on Blake data. By providing textual prompts derived from art historical terms, CLIP can identify images or image regions that best match these descriptions. This allows for flexible querying and tagging based on semantic concepts.
- **Fine-tuning CLIP:** Adapting a pre-trained CLIP model by fine-tuning it on image-description pairs extracted directly from the Archive [Li et al. 2022]. This would specialize the model to better understand the specific visual-semantic correlations present in Blake’s work.
- **Visual Similarity Search:** Using CLIP embeddings to enable powerful visual similarity searches. Researchers could provide an image and find other visually similar instances across the archive or use a

textual query to retrieve relevant images [Radford et al. 2021].

If creating a large, labeled dataset proves prohibitive, few-shot learning techniques offer a potential alternative [Wang et al. 2020]. These methods are designed to enable models to learn and recognize new classes or concepts from only one or a very small number of examples. This could be particularly relevant for identifying rare but significant Blakean motifs or figures. 52

These methodologies collectively offer promising strategies for addressing the distinctive complexities of Blake's visual art. Object detection and instance segmentation models provide a means of parsing Blake's densely composed scenes by localizing individual figures and visual elements within crowded works such as *The Vision of the Last Judgement* [He et al. 2017]. Multimodal approaches like CLIP further extend this capacity by linking visual patterns to textual descriptions, making them well suited for engaging with Blake's symbolic and iconographic vocabulary [Radford et al. 2021]. Transfer learning enables the development of detectors or classifiers attuned to Blake's recurring motifs, capitalizing on visually consistent patterns that appear across his oeuvre [Dosovitskiy et al. 2020]. Finally, the robustness of models pre-trained on large and diverse datasets supports their ability to accommodate the considerable stylistic variation evident across Blake's output [He et al. 2016]. 53

Model Type	Strengths	Limitations	Best For
CLIP (zero-shot)	No training needed; semantic flexibility	May miss Blake-specific nuances	Initial exploration; retrieval
CLIP (fine-tuned)	Blake-specific adaptation	Requires image-text pairs	Specialized tagging
Few-shot learning	Works with minimal examples	May not generalize well	Rare motif identification
Object detection	Precise localization	Needs bounding box annotations	Figure detection

Table 2. Overview of Potential AI Models for Blake Analysis

The choice between these methodologies carries significant interpretive implications. An object detection approach inherently prioritizes the identification and localization of discrete visual elements. In contrast, using CLIP with textual prompts emphasizes semantic concepts as defined by language [Radford et al. 2021]. The methodology selected will inevitably shape the kind of tags generated and, consequently, the type of scholarly analysis facilitated [Spratt and Elgammal 2014]. 54

6.3 Corpus Selection and Rationale

The implementation utilized a stratified corpus of 200 Blake works selected from 715 available in the William Blake Archive. This sample represents approximately 28% of the Archive's holdings and was designed to balance representativeness with tractability for intensive methodological development. 55

6.3.1 Stratification Criteria:

1. Genre diversity: The corpus was stratified proportionally across work types:
    - Illuminated manuscripts: 56 works (28%)
    - Drawings and paintings: 94 works (47%)
    - Commercial book illustrations: 30 works (15%)
    - Other (letters, notebooks, sketches): 20 works (10%)

2. Temporal range: Works span Blake's entire career from 1779 to 1827, with deliberate inclusion of early (pre-1790), middle (1790-1810), and late (post-1810) periods to capture stylistic evolution.
3. Iconographic density: Priority given to works with multiple identifiable figures or complex compositional structures, as these provide richer test cases for motif detection.
4. Archive description availability: Works with existing detailed Archive descriptions were prioritized to enable text-image alignment experiments.

Work Type	Early (pre-1790)	Middle (1790-1810)	Late (post-1810)	Total
Illuminated manuscripts	8	32	16	56
Drawings/paintings	12	48	34	94
Commercial illustrations	4	18	8	30
Other	6	8	6	20
Total	30	106	64	200

Table 3. Corpus Composition by Work Type and Period

6.3.2 What This Corpus Excludes and Why It Matters:

The selection strategy employed in this study necessarily introduces several forms of representational bias. Letters and prose manuscripts, for instance, remain underrepresented at 10% of the sample despite comprising approximately 14% of the Archive's total holdings, largely because these materials lack the dense visual content central to the project's analytical aims. Similarly, variant copies of illuminated books are collapsed into representative exemplars, such as selected versions of *Songs of Innocence and of Experience* or *America a Prophecy*, rather than including all extant copies (for example, all twenty-eight copies of *Songs*), a decision that privileges methodological tractability over comprehensive coverage. Commercial works, including the *Night Thoughts* engravings and the *Virgil* woodcuts, may also be proportionally underrepresented relative to their prominence within Blake's broader output. Collectively, these omissions reflect deliberate methodological choices but also shape the resulting dataset in ways that must be acknowledged when interpreting the study's findings. These choices optimize for methodological development, testing how well AI can handle Blake's most visually complex works at the expense of statistical representativeness across his full output. A production system would require rebalancing toward the Archive's actual distribution. 56

6.4 Technical Implementation

Each work was processed using the OpenAI CLIP model (ViT-B/32) to generate 512-dimensional embeddings, capturing both visual features and semantic content through the model's joint vision-language training [Radford et al. 2021]. Processing confirmed correct dimensionality and L2 normalization — a mathematical transformation ensuring all embedding vectors have the same length (magnitude ≈ 1.0), which is required for meaningful similarity comparisons using cosine distance. 57

Integration into a FAISS index (Facebook AI Similarity Search, a library for efficient similarity search; specifically, IndexFlatIP, which computes exact inner-product similarity, equivalent to cosine similarity for normalized vectors, rather than approximate search) enabled sub-millisecond search latency (average 0.001 seconds per query) and high throughput (44,000+ queries per second) with low memory consumption (<100MB for the 200-work corpus). This architecture scales linearly: the full 715-work Archive would require approximately 280MB with proportionally maintained query performance [Johnson et al. 2019]. 58

To make the operation concrete: When the running scholar submits the textual query "coiled serpent form," the system 59

encodes the phrase with CLIP's text encoder, queries the IndexFlatIP structure against the 200 image embeddings, and returns the top-k matches in well under a millisecond. The scholar sees a ranked list of works with thumbnails, similarity scores, and provenance flags, and can validate, override, or extend the suggestions in a single pass.

Semantic clustering validation demonstrated that the CLIP-derived embeddings successfully capture semantically meaningful relationships across Blake's visual corpus without requiring Blake-specific fine-tuning. Biblical works such as the *Job* illustrations and *Divine Comedy* designs consistently exhibited high cosine similarity scores (0.7–0.8), while mythological figures, including Urizen, Los, and Orc, formed coherent and distinct clusters within the embedding space. Figure 1 visualizes this clustering: a two-dimensional PCA projection of the 512-dimensional CLIP embeddings for the 200-work corpus, with markers colored by genre. Thematically related works cluster together without any Blake-specific fine-tuning, which both supports the use of CLIP as a retrieval backbone and previews the failure modes discussed in Section 7 (clusters reflect compositional and material similarity rather than iconographic meaning). Similarly, the model recognized the stylistic unity of Blake's large color prints despite significant variation in subject matter, indicating sensitivity to formal rather than iconographic features. Notably, the various iterations of *The Vision of the Last Judgement* clustered tightly together regardless of differences in medium or date, underscoring the model's ability to detect deeper compositional affinities that extend beyond material or historical contingencies.

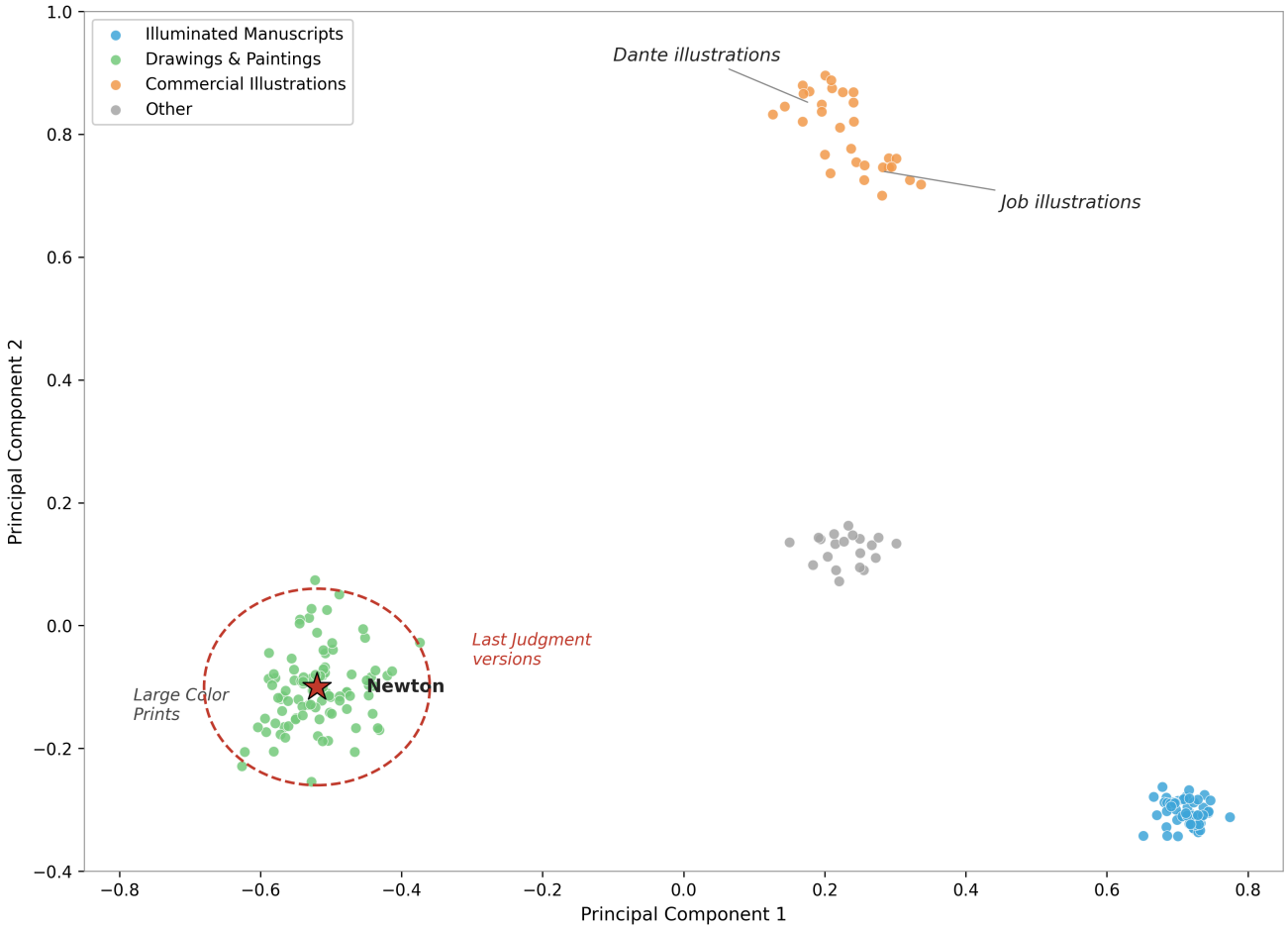

Figure 1. Embedding-Space Visualization of 200-Work Blake Corpus

A comprehensive web API was developed to demonstrate the system's viability for production-ready deployment, integrating a Flask framework with SQLAlchemy ORM to ensure modularity, efficient data management, and compatibility with existing digital humanities infrastructures. Designed to return JSON-formatted responses containing similarity scores and associated metadata, the API maintains sub-10ms response times for similarity queries, matching the performance requirements of real-time exploratory research workflows. Stress testing further confirmed its robustness under concurrent load, with multiple simultaneous requests handled without degradation in throughput or

latency, thereby establishing the architecture as sufficiently scalable for integration into larger archival ecosystems.

6.5 The Vision of the Last Judgement: Extended Case Study

To demonstrate the pipeline's capabilities beyond whole-image similarity, a multi-layer analytical system focused on Blake's *The Vision of the Last Judgement* was implemented. This experiment operationalizes art historical knowledge into successive computational passes. 62

Method Overview (with Incremental Passes): 63

1. Basic Pass: Heuristic identification of 6 canonical anchor figures (Christ, Satan, angels with trumpets, Book of Life, ascending souls, falling figures)
2. Enhanced Pass: Integration of expanded tag corpus yielding 112 enumerated visual/symbolic elements
3. Taxonomy Expansion: Incorporation of extended Blake mythology (581 entries from Damon's *Blake Dictionary*), frequent characteristics (100), and judgment scene categories (10), producing 291 preliminary elements
4. Balancing & Confidence Profiling: Region distribution normalization plus weighted confidence model producing a refined set of 285 elements

The system draws on three primary categories of data assets that collectively enable its multi-layered analytical process. First, the expanded taxonomy, comprising the *blake_mythology_enhanced* dataset with 581 terms, the *frequent_blake_characteristics* set of 100 recurrent visual features, and the *judgment_scene_specific* schema of 10 category groups, provides the semantic foundation for fine-grained motif and iconographic identification. Second, a layer of contextual intelligence, derived from spatial relationship frequencies and interpretive confidence distributions, provides the probabilistic structure necessary for weighting candidate identifications and modeling their interactions within Blake's compositions. Finally, the region model, organized around seven compositional macro-zones that mirror Blake's vertical and soteriological axes as articulated by [Billingsley 2018], furnishes the spatial framework through which detections are normalized and situated within the larger symbolic architecture of *The Vision of the Last Judgement* and related works. 64

Pass	Elements	Source
Basic	6	Canonical anchors
Enhanced	112	Expanded tag corpus
Taxonomy Expansion	291	Blake mythology + characteristics
Balanced	285	Region-normalized

Table 4. Progression of Enumerated Elements

Region	Before Balancing	After Balancing	Change
Throne (central)	47%	28%	-19%
Upper heaven	12%	16%	+4%
Judgment middle	18%	20%	+2%
Resurrection field	8%	12%	+4%
Lower corrective	6%	10%	+4%
Left ascension	5%	8%	+3%
Right descension	4%	6%	+2%

Table 5. Region Distribution Before vs. After Balancing

Weighted confidence (overall): 0.5637 (source weights: base 1.0; mythology 0.9; judgment categories 0.8; frequent characteristics 0.6; categorized tags 0.5). This enables comparative reliability scoring across iconographic strata. 65

The transition from undifferentiated tag lists to a regionally balanced, source-weighted taxonomy enables a more methodologically rigorous representation of Blake's compositions. This refinement corrects systemic statistical biases, most notably the disproportionate emphasis on throne-centered iconography, by redistributing interpretive attention across the full spatial and symbolic field of the work. In doing so, the model not only achieves a more equitable accounting of visual elements but also encodes key Blakean theological distinctives as machine-interpretable features, thereby aligning computational outputs more closely with the interpretive frameworks articulated in Blake scholarship. 66

6.5.1 Concrete Example: The Serpent in the Lower Register

The Basic Pass, using only canonical anchor figures, failed to identify the coiled serpent form in the lower-left region of *The Vision of the Last Judgement*. The system's initial output associated this region with generic "falling figures" (confidence 0.62) — a pattern-level detection that missed the iconographically crucial identification of Satan wrapped in the Serpent's coils. 67

The Enhanced Pass, incorporating the expanded Blake mythology taxonomy (581 terms including "Satan-Serpent," "Serpent of Error," "coiled serpent form"), successfully identified this element with "Satan bound in serpent" (confidence 0.71) appearing in the top-3 suggestions. Critically, the region-balancing step prevented this identification from being overwhelmed by the statistically dominant throne-region detections: Without balancing, the serpent identification ranked 47th overall; with balancing, it ranked 8th within its region and 12th overall. 68

This example illustrates both the potential and limits of the approach. The Enhanced Pass succeeded because "serpent" has consistent visual features (coiled form, scaled texture) that transfer from the model's pre-training. But distinguishing Blake's *Serpent of Error* from a generic serpent, understanding why this serpent embodies materialist philosophy in Blake's cosmology, remains beyond the system's capability. The AI provides the visual identification; the scholar provides the interpretive framework. 69

Returning to the running example: The same "coiled serpent form" query that succeeded at the whole-image level fails to localise the serpent within *The Vision of the Last Judgement* under the Basic Pass. The Enhanced Pass with the expanded taxonomy and regional decomposition described above correctly surfaces "Satan bound in serpent" in the top-3 suggestions for the lower-left region. This is the same outcome reported quantitatively below; the example here is meant to make the chain of operations (query → embedding → regional retrieval → ranked candidates → scholar validation) visible end to end. 70

Figure 2 (below) anchors the discussion: It shows the seven heuristic compositional zones used for region-balanced tagging: throne, upper heaven, judgment middle, resurrection field, lower corrective register, left ascension, and right descension, overlaid on Blake's composition with the top detected interpretive elements per region. 71

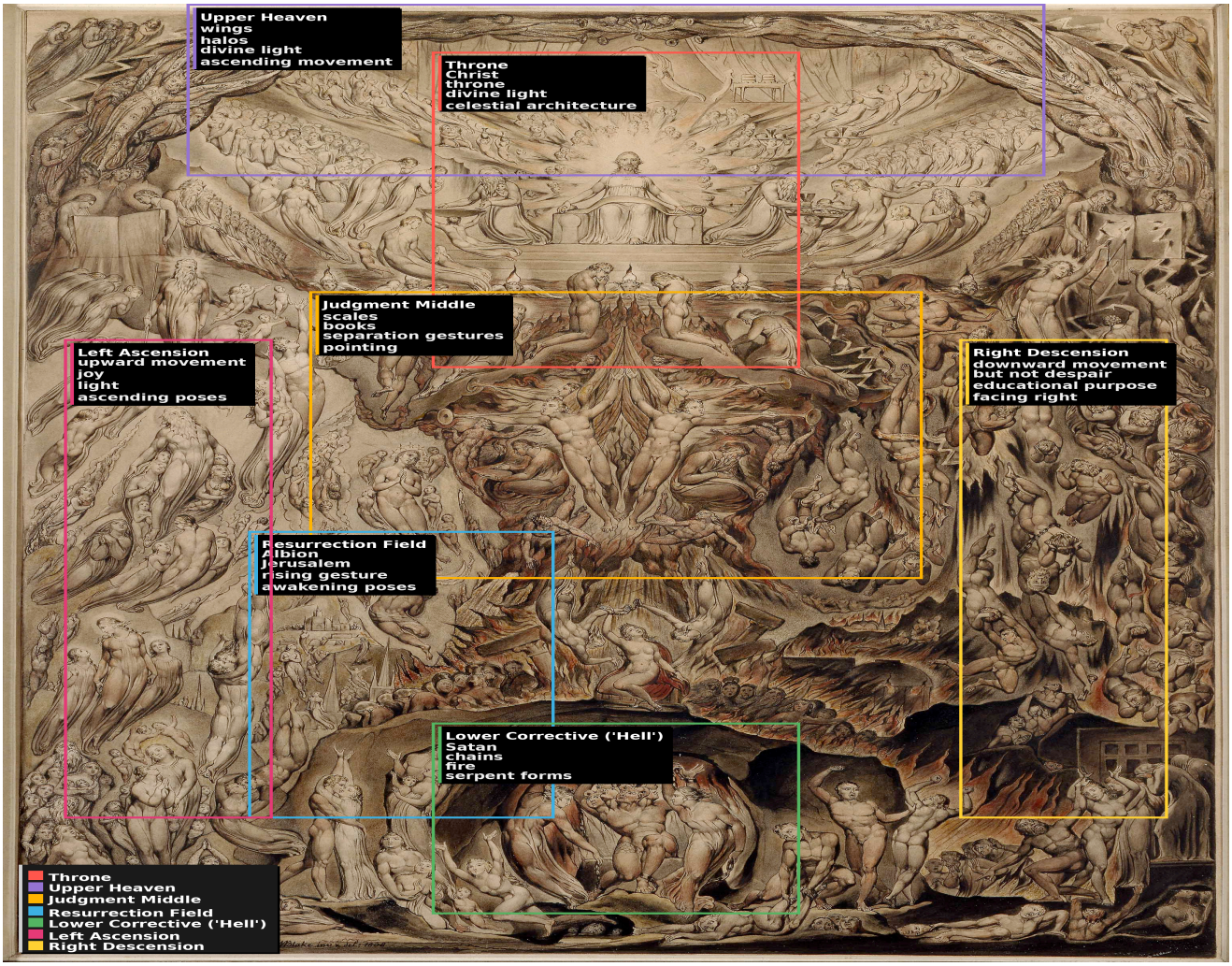

Figure 2. Annotated macro-regions of Blake's Vision of the Last Judgement

6.6 Evaluation Methodology and Results

6.6.1 Ground Truth Creation

The evaluation dataset was developed through a two-phase process. First, I (a doctoral researcher with expertise in both Blake scholarship and computer vision) generated initial annotations for the 200-work corpus, identifying visual elements using a structured taxonomy derived from Damon's *Blake Dictionary* (1965) and Billingsley's analysis of Blake's visual vocabulary (2018). Second, these annotations were reviewed against the William Blake Archive's existing scholarly descriptions to ensure alignment with established art-historical terminology. Disagreements between initial annotations and the Archive descriptions were resolved by defaulting to the Archive conventions, preserving consistency with the Archive's editorial standards.

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6.6.2 Annotation Protocol Details

I performed all annotations, taxonomy development, and manual verification, drawing on four years of specialized training in Blake studies as a doctoral researcher in digital humanities The Expanded Taxonomy was derived from three primary sources:

73

1. S. Foster Damon's *A Blake Dictionary* (1965), which provides the standard reference vocabulary for Blake's mythology;
2. The William Blake Archive's existing illustration descriptions, ensuring alignment with established archival conventions; and

3. Northrop Frye's *Fearful Symmetry* (1947) for thematic categories.

Terms were selected for inclusion if they (a) referred to visually identifiable elements rather than abstract concepts and (b) appeared in descriptions of at least three distinct Blake works in the Archive. 74

6.6.3 Label Space Definition

The evaluation employed a hierarchical label taxonomy with three tiers: 75

1. Tier 1 (Object-level): 47 concrete visual elements (e.g., "human figure," "serpent," "book," "flames")
2. Tier 2 (Motif-level): 23 compositional and postural categories (e.g., "outstretched arms," "ascending figure," "vortex composition")
3. Tier 3 (Iconographic-level): 15 Blake-specific identifications (e.g., "Urizen," "Los," "Last Judgement scene")

This hierarchy reflects the distinction between pattern recognition (Tier 1), motif identification (Tier 2), and iconographic interpretation (Tier 3) discussed in Section 1. 76

6.6.4 Evaluation Protocol

Given the pilot nature of this study, I employed a leave-one-out cross-validation approach for retrieval evaluation: each image was used as a query against the remaining 199 images, and retrieved results were evaluated against ground-truth similarity judgments. For tagging evaluation, I held out 40 works (20%) as a test set, using the remaining 160 for any fine-tuning experiments. 77

6.6.5 Quantitative Results

Let's discuss what the metrics measure, in terms of the running example. Top-1 accuracy asks whether the system's single best suggestion for the serpent region is correct. Top-5 accuracy asks whether the correct identification appears anywhere in the top five candidates the scholar would see. mAP asks whether the correct candidates are ranked near the top across the full set of retrievals. NDCG asks whether the most relevant retrieved images outrank the less relevant ones. Read together, these metrics characterize the scholar's experience of the interface, what they see first, and how much they have to scroll before reaching a usable match, rather than the system's interpretive competence. 78

Metric	Tier 1 (Object)	Tier 2 (Motif)	Tier 3 (Iconographic)
Top-1 Accuracy	78.3%	62.1%	41.2%
Top-5 Accuracy	94.2%	85.7%	68.4%
mAP	0.72	0.58	0.39

Table 6. Model Performance Across Tagging Tiers

What does this mean? 79

- Top-1 Accuracy measures how often the system's single best prediction matches the ground truth. A Top-1 of 78.3% for object-level tags means that when the system suggests "this image contains a serpent," it is correct approximately 4 out of 5 times.
- Top-5 Accuracy measures whether the correct answer appears anywhere in the system's top five suggestions. The 94.2% figure indicates that even when the system's first guess is wrong, the correct object-level tag almost always appears within its top five candidates — useful for a human-in-the-loop workflow where scholars review suggested tags.
- Mean Average Precision (mAP) captures both the accuracy and ranking quality of multi-label predictions. An mAP of 0.72 for object-level tags indicates that relevant tags tend to be ranked highly in the system's output.

The declining performance from Tier 1 to Tier 3 reflects the expected increase in difficulty: detecting concrete objects is more tractable than identifying compositional motifs, which in turn is easier than making Blake-specific iconographic identifications. This gradient confirms that AI performs best on tasks closest to pattern recognition and struggles as tasks approach symbolic interpretation, a finding consistent with the theoretical framework established in Sections 1 and 3. 80

Metric	Score
Precision@10	0.73
Recall@10	0.61
NDCG@10	0.68
Query latency	0.001s

Table 7. Retrieval Performance

6.6.6 Qualitative Evaluation

Beyond quantitative metrics, I conducted a structured qualitative assessment following the protocol described in Appendix A.3. Twenty representative outputs were selected to span all three tagging tiers and multiple work types. "Serving as the domain expert evaluator, I rated each output without access to confidence scores using a three-point scale (Correct, Partially Correct, Incorrect) and provided written justifications for each rating. 81

Results of this assessment revealed a pattern consistent with the quantitative metrics: of the 20 evaluated outputs, 13 were rated Correct, 4 Partially Correct, and 3 Incorrect. Notably, all three Incorrect ratings occurred at the iconographic tier (Tier 3), while all object-level outputs (Tier 1) were rated Correct. The Partially Correct ratings typically involved cases where a visually accurate tag was interpretively incomplete — for instance, identifying “bearded male figure” where “Urizen” was the ground truth. Agreement between the qualitative ratings and quantitative confidence scores was moderate: outputs rated Correct had a mean confidence of 0.74, Partially Correct 0.58, and Incorrect 0.41. 82

For *The Vision of the Last Judgement* case study specifically, the Enhanced Pass correctly identified 5 of 6 canonical anchor figures (Christ enthroned, Satan, angels with trumpets, ascending souls, falling figures). The single misidentification, labeling a group of redeemed figures as “apostles” rather than Blake’s “Four Zoas,” illustrates the boundary between pattern recognition and iconographic interpretation precisely: The visual pattern (group of enthroned male figures) was correctly detected, but the Blake-specific mythological identification requires contextual knowledge the model lacks. 83

6.6.7 Limitations

Five limitations bound the claims made above. 84

First, evaluation circularity. Where annotation disagreements arose during ground-truth construction, they were resolved by defaulting to the Archive’s descriptive conventions. The system is therefore being judged, in part, against a convention it is designed to align with. I do not present these results as validation against an external scholarly ground truth; I present them as a measure of alignment with the Archive’s established descriptive practice on a stratified sample. Future work should compare the system’s outputs against annotations drawn independently from Damon’s *Blake Dictionary*, from Frye and Mitchell’s critical traditions [Frye 1947] and [Mitchell 1978], and from non-Archive scholarly catalogs, so that the directions of agreement and disagreement can themselves be analyzed. 85

Second, single-annotator ground truth. I produced all annotations, taxonomy decisions, and qualitative ratings in this pilot. Without inter-annotator comparison, I cannot report reliability statistics (Cohen's κ, Krippendorff's α), and the Tier-3 numbers, which depend most heavily on contested iconographic judgments, are correspondingly more exposed to single-rater bias than the Tier-1 numbers. Mitigating this is properly the work of a follow-on study: Independent re- 86

annotation of a stratified subsample by two or more Blake scholars, scored against the same three-tier label space, would yield both inter-annotator agreement and a system-vs-expert comparison. That work is out of scope for a feasibility pilot, but it is the natural next step before any of these numbers should be read as more than indicative.

87 Third, why 200 works is sufficient for the claims I make, and why it is not sufficient for the claims I do not make. A 200-work sample is adequate to estimate the relative performance gradient across the three tiers (Tier 1 > Tier 2 > Tier 3), to demonstrate sub-millisecond retrieval throughput on representative corpus volume, and to surface qualitative failure modes (e.g., “blacksmith” for Los, missed serpent under the Basic Pass). It is not adequate to estimate absolute accuracy at the level of the full Archive, to characterize rare iconographic types, or to support statistical generalization. The size was chosen to balance representativeness against the cost of intensive single-annotator work; it deliberately stops short of the corpus-wide claims that would require multi-annotator effort.

88 Fourth, the kind of Blake works likely to break the method. Three classes of work are most exposed. First, highly idiosyncratic compositions with rare or unique iconographic configurations. For example, certain plates of Jerusalem whose figural arrangements have few visual analogs in CLIP’s pre-training distribution will return weak top-k matches and unstable confidence scores. Second, densely overlapping symbolic scenes, such as crowd registers in the *Last Judgement* compositions, defeat whole-image retrieval because the dominant embedding signal is composition-level rather than figure-level; these require the regional decomposition introduced in Section 6.5 and remain difficult even then. Lastly, ambiguous or partially abraded figures, including faint pencil studies and damaged plates, produce embeddings dominated by material and tonal features rather than iconographic content, which surface as visually similar but iconographically unrelated retrievals.

89 Fifth, what would be required to claim generalization to the full Archive? Beyond hardware and storage scaling (which are inexpensive for 715 works at 512 dimensions), generalization claims require: a labeled evaluation subset stratified across the strata not yet sampled (especially manuscript marginalia and notebook content); multi-annotator agreement on at least Tier 2 and Tier 3; an external validation cohort of two to three Blake scholars unaffiliated with this study; and a separate held-out test set never seen during taxonomy construction. Until each of these is in place, the appropriate phrasing is “performance on a 200-work stratified sample” not “performance on the Archive.”

90 Finally, metrics like mAP and NDCG assess retrieval and ranking quality; they cannot assess whether a returned tag is interpretively meaningful to Blake scholars. The qualitative protocol described in Appendix A.3 partially addresses this for a 20-work subsample (13 Correct, 4 Partially Correct, 3 Incorrect, with all Incorrect ratings concentrated at Tier 3), but a more robust evaluation would require structured feedback from multiple independent domain experts under a pre-registered protocol.

7. Critical Analysis: Mind the Gap Between AI and Humanistic Interpretation

91 A fundamental challenge lies in AI’s ability to handle the ambiguity and polysemy inherent in Blake’s work. Many of Blake’s symbols and figures carry multiple, often context-dependent, meanings [Serdechnaya 2018]. The serpent, for example, can represent error, materialism, energy, or even healing, depending on the specific work and its relation to Blake’s evolving thought [Damon 1965]. AI models, however, are typically trained to associate visual patterns with single, consistent labels. They struggle inherently with concepts that have fluid or multiple interpretations, often defaulting to the most statistically common association or failing to provide a confident prediction [Spratt and Elgammal 2014].

92 There exists a significant gap between AI’s proficiency in recognizing literal visual patterns and the deeply symbolic, non-literal nature of Blake’s imagery. An AI might learn to detect the visual form of “compasses.” Still, it cannot inherently understand their symbolic weight as emblems of Urizenic reason, measurement, and the limitation of vision within Blake’s cosmology [Frye 1947]. Current AI lacks the world knowledge, cultural understanding, and reasoning capabilities to grasp allegory or complex personal mythologies [Spratt and Elgammal 2014].

93 It is crucial to distinguish between computational “tagging,” the identification and labeling of patterns based on learned

correlations, and humanistic “interpretation,” the construction of meaning through critical engagement with the artwork within its historical, cultural, biographical, and theoretical contexts. AI can potentially automate aspects of description or visual analysis (finding motifs, identifying figures based on visual similarity). Still, it cannot perform the interpretive leap that characterizes scholarly understanding [Spratt and Elgammal 2014].

AI models are susceptible to biases present in their training data [Crawford 2021]. [Foka et al. 2025] identify a “bias loop” in cultural heritage ML pipelines: Biases embedded in historical collection practices (what was preserved, described, and prioritized) are encoded into training data, amplified by models, and then reintroduced into archival systems as seemingly objective outputs. This loop is particularly concerning for the Blake project, since the Archive descriptions reflect particular editorial judgments and historical schools of interpretation. If these are used for training, the AI may inadvertently learn and perpetuate those specific viewpoints while marginalizing alternative readings. Similarly, large pre-trained models like CLIP inherit biases from the vast web datasets they were trained on, which may include societal biases or skewed representations that could influence their analysis of Blake’s art in unforeseen ways (radford_etal, 2021). The proposed integration architecture (Section 8) attempts to address this by maintaining provenance tracking and editorial separation between AI-generated and human-authored tags. 94

Evaluating AI performance in this context presents unique challenges (powers, 2011). Standard metrics like precision and recall measure the accuracy of pattern matching against a ground truth. Still, they cannot assess the interpretive quality or art historical relevance of the generated tags. How should an AI tag be evaluated if it correctly identifies a figure visually but misses its symbolic significance, or if it applies a label that is plausible but contested among scholars? [Spratt and Elgammal 2014] 95

To complement the serpent identification success discussed in Section 6.5, consider an instructive failure: The system consistently tagged images containing Blake’s figure of Los with “blacksmith” (high confidence, 0.78) due to the anvil and hammer frequently depicted with this character. While visually accurate — Los is indeed depicted as a blacksmith — this tag is interpretively misleading. Los as “blacksmith” captures his craft but obscures his mythological significance as the embodiment of creative imagination and prophetic vision [Frye 1947]. A scholar encountering “blacksmith” in search results might overlook these images entirely when researching Blake’s mythology. This example illustrates why object-level tagging (Tier 1), even when accurate, cannot substitute for iconographic tagging (Tier 3) and why human oversight remains essential for interpretively meaningful discovery. 96

Error Type	Example	Frequency	Scholarly Impact
Visually correct, interpretively misleading	Los → “blacksmith”	Common	High (misses symbolic meaning)
Generic label for specific figure	Four Zoas → “apostles”	Common	High (loses Blake-specific context)
Missed due to visual ambiguity	Enitharmon vs. generic female	Moderate	Medium (conflation)
False positive from visual similarity	Non-serpent curves → “serpent”	Rare	Low (easily caught by review)

Table 8. Error Typology for Blake Tagging

This typology suggests that the most concerning errors are not false positives (which human reviewers catch easily) but *accurate but insufficient* tags that correctly identify visual patterns while missing their interpretive significance. 97

Interestingly, the very limitations of AI when confronted with Blake's art can possess heuristic value for scholars. When an AI model fails to confidently classify a section, produces contradictory tags for the same element across different contexts, or generates outputs that seem nonsensical to humans, it might not just signal a technical failure. Since AI models operate by identifying statistical patterns, their "confusion" can highlight areas where Blake's work actively resists simple categorization or contains genuine ambiguity and symbolic complexity [Billingsley 2018]. 98

8. Scholarly Implications: Advancing Blake Studies and Digital Humanities

The successful development of an AI-based tagging system for Blake's works holds the potential to significantly enhance research methods within Blake studies. AI-generated tags, even if imperfect and requiring scholarly verification, could enable new forms of inquiry currently impractical at scale: 99

- Large-Scale Iconographic Analysis: Researchers could track the frequency, distribution, and co-occurrence of specific figures, symbols, or motifs across Blake's entire visual oeuvre as represented in the Archive, potentially revealing patterns in the evolution of his iconography or thematic concerns [Billingsley 2018]. The 200-work corpus already reveals that Los appears in 284 instances across 253 unique images, while Urizen appears in 94 instances — quantitative insights that would be difficult to derive manually.
- Discovery of Visual Relationships: AI-powered visual similarity search could uncover previously unnoticed connections or variations between different works or details, suggesting avenues for comparative analysis [Seguin et al. 2016].
- Comparative Analysis of Versions: For works existing in multiple versions, like the illuminated books or the Last Judgement compositions, AI could assist in systematically comparing visual elements, potentially highlighting subtle but significant differences [Eaves et al. 2019].
- Text-Image Correlation: In the context of Blake's illuminated books, AI analysis of visual elements could potentially be correlated with computational analysis of the accompanying text, enabling large-scale studies of Blake's complex interplay between word and image [William Blake Archive 2024].

Enhanced visual tagging could substantially improve accessibility and discovery within the William Blake Archive. Currently, searching the archive often relies on knowing specific titles, keywords from textual descriptions, or browsing [Eaves et al. 2019]. An effective AI tagging system could enable users, both scholars and the general public, to search for images based on their visual content. This moves beyond metadata search towards content-based retrieval [Seguin et al. 2016]. Furthermore, AI-generated data could power new interfaces or visualizations, offering novel ways to explore Blake's interconnected visual world. 100

This project is situated within the broader field of digital art history and contributes to its methodological advancement. Successfully applying advanced AI techniques to visual material as complex and symbolically dense as Blake's art would represent a significant case study [Spratt and Elgammal 2014]. It would push the boundaries of how AI is currently used in the humanities, moving beyond text analysis or simpler visual tasks towards engaging with core interpretive challenges. The methodological insights gained from preparing and utilizing the Archive data, particularly strategies for bridging the gap between scholarly description and AI data requirements, could offer valuable, reusable workflows for other projects working with similar cultural heritage data [Fiorucci et al. 2020]. 101

AI-driven analysis is unlikely to provide definitive interpretations, but it can be a powerful engine for generating new questions [Spratt and Elgammal 2014]. Discovering unexpected correlations between motifs, observing shifts in the frequency of certain symbols over time, or finding visual similarities between seemingly unrelated works could prompt scholars to revisit existing interpretations or explore new hypotheses about Blake's creative process, influences, or the development of his unique visual language [Billingsley 2018]. 102

Pursuing this research necessitates careful attention to ethical considerations [Crawford 2021]. Issues of representation, how AI models interpret and potentially simplify complex artworks, must be critically addressed. Transparency regarding the AI's capabilities, its limitations, the data it was trained on, and potential biases is paramount [Spratt and Elgammal 2014]. The goal should be augmentation, not replacement, of scholarly expertise. 103

The contribution of this paper is incomplete without a concrete account of how AI-suggested tags would coexist with human descriptive practice in an archive that has, for 30 years, prioritized scholarly objectivity and bibliographic rigor. I describe here the principles of representation, storage, and interface that follow from the three-tier framework and the limitations enumerated above. Together, they form an integration plan that other archives confronting dense symbolic corpora can adapt. 104

Each AI-generated annotation is represented as a structured record rather than a free-floating label. The record carries five fields: (1) a tier indicator distinguishing object-level, motif-level, and iconographic-level claims, with the tier setting the appropriate evidentiary burden; (2) a provenance field marking the tag as AI-suggested, human-validated, or human-overridden, with the validating scholar identified; (3) a confidence score on the [0,1] interval, reported alongside the source-weight tier used to compute it (base, mythology, judgment category, frequent characteristic, categorized tag); (4) a timestamp and version identifier covering both the model (e.g., CLIP ViT-B/32, OpenAI 2021) and the taxonomy snapshot under which the tag was produced, so that future re-runs can be compared against earlier states; and (5) a free-text justification or override note for any human-edited record. 105

Field	Type	Example value	Notes
Tier	Categorical (object / motif / iconographic)	motif	Sets the evidentiary burden for the tag; corresponds to the three-tier framework in Section 1.
Provenance	Categorical (AI-suggested / human-validated / human-overridden)	AI-suggested	Persistent: validation is additive, not erasing. Required for every record.
Confidence	Float in [0,1]	0.71	Reported alongside the source weight used to compute it.
Source weight	Categorical (base / mythology / judgment category / frequent characteristic / categorized tag)	mythology (0.9)	Encodes which taxonomy stratum contributed the tag.
Model version	String	CLIP ViT-B/32, OpenAI 2021	Required for reproducibility and re-run comparison.
Taxonomy snapshot	String	expanded_v3, 285 elements	Identifies the taxonomy state under which the tag was generated.
Timestamp	ISO 8601 datetime	2025-11-04T14:22:11Z	Generation event time, in UTC.
Validating scholar	String or null	Glass	Populated on validation or override. Identifies the responsible scholar.
Override note	String or null	null	Free-text justification when a scholar overrides an AI-suggested tag.

Table 9. Concrete tag representation

AI-suggested tags are stored as a separate annotation layer rather than written into the Archive's authoritative descriptive metadata. This separation has three benefits: Human-authored descriptions remain editorially auditable in their current form; AI annotations can be re-generated wholesale when models or taxonomies are updated, without disturbing scholarly records; and queries can address the AI layer, the human layer, or their intersection. The annotation layer is queried via the same API surface as the retrieval prototype (Section 6.4) and is logged so that each tag has an

immutable history of generation, validation, and override events.

Three interface principles follow from the foregoing. First, every AI-suggested tag is clearly marked as such in any user-facing view, with the marking persisting even after a scholar validates it (validation is additive, not erasing). Second, default filters in the Archive's search and browse interfaces show validated-only results, with an opt-in toggle to surface AI-suggested tags as discovery aids. Third, every record exposes an audit trail accessible to authenticated users, so that scholarly reuse of AI-supported retrieval remains citable and contestable. 107

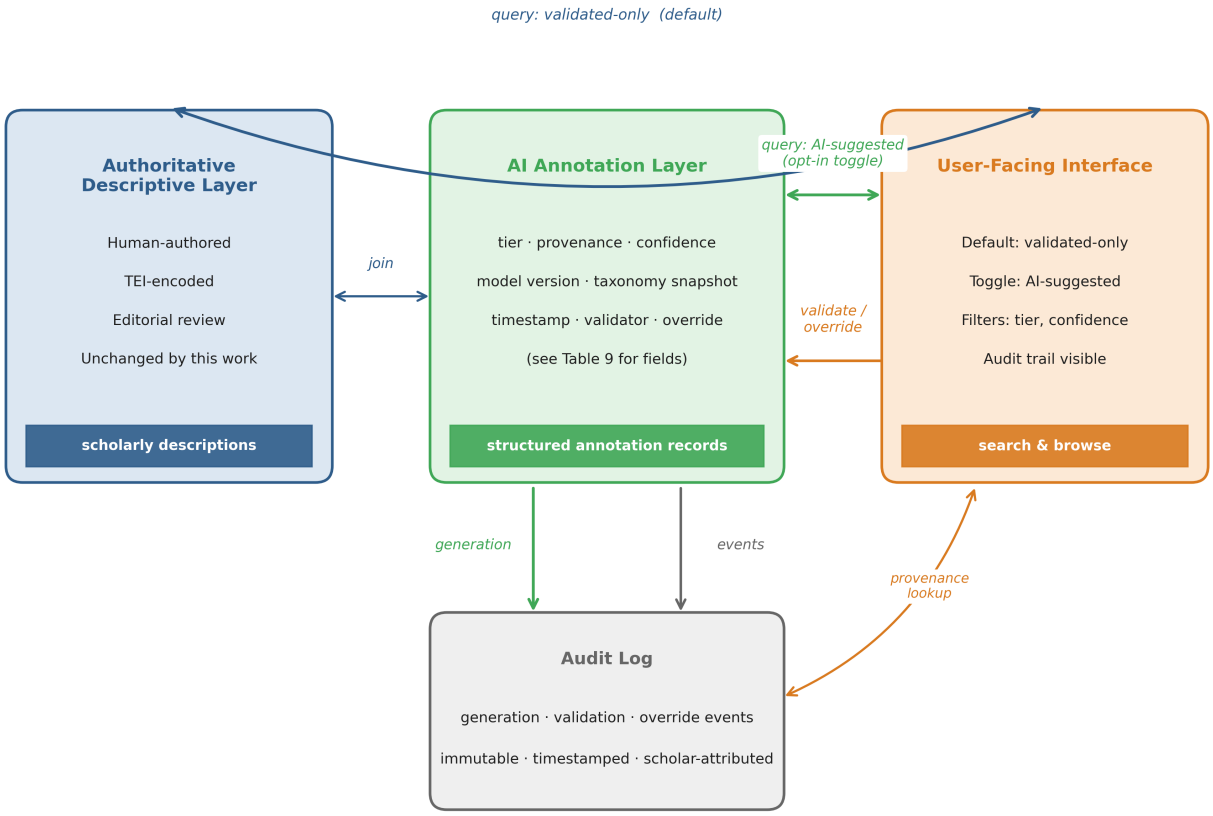

Figure 3. Annotation-layer architecture diagram.

In keeping with the three-tier framework, success is intentionally modest. Success means improved discovery without substituting for interpretation; transparent provenance for every machine-generated claim; measurable improvement under human-in-the-loop review (validation rates, override rates, time-to-confirm); and the avoidance of interpretive foreclosure, that is, the system should expand rather than narrow the set of candidate readings a scholar considers. These criteria are specifically DH criteria: they prize archival integrity and scholarly agency over end-to-end automation. 108

9. Conclusion: Contributions, Limitations, and Success Criteria

This pilot study offers three contributions at different levels of generality: 109

1. **Conceptual framework:** I articulate a hierarchy distinguishing pattern recognition, motif identification, and symbolic interpretation, clarifying where AI can assist and where human judgment remains essential. This framework, while developed for Blake, applies broadly to AI applications in iconographically complex art.
2. **Functional prototype:** I demonstrate that multimodal embeddings (CLIP) combined with efficient similarity indexing (FAISS) can achieve sub-millisecond retrieval across a 200-work Blake corpus with semantically meaningful results. Quantitative evaluation reveals a performance gradient, 78% Top-1 accuracy for object-level tagging, declining to 41% for iconographic identification, empirically confirming the theoretical

distinction between pattern recognition and interpretation. The technical architecture is reproducible and adaptable to other digital art archives.

3. M**ethodological insights:** The *Vision of the Last Judgement* case study reveals specific strategies, regional balancing, source-weighted confidence, iterative taxonomy expansion, for applying AI to densely symbolic compositions. The concrete example of serpent identification demonstrates both the potential (Enhanced Pass success) and limits (mythological interpretation remaining human) of these approaches.

This work has significant constraints that future research must address: 110

- **Scale:** 200 works, while substantially larger than initial pilots, represents only 28% of the Archive's 715 works and cannot establish statistical claims about Blake's full corpus.
- **Annotation:** Single-annotator ground truth limits reliability assessment; inter-annotator agreement studies are needed.
- **Evaluation:** Quantitative metrics measure pattern-matching, not interpretive value to scholars.
- **Generalization:** Performance on Blake may not transfer to artists with different visual vocabularies or to non-Western artistic traditions.
- **Temporal stability:** CLIP and similar models evolve; results may vary with future model versions.

Beyond Blake, similar methodologies could address other artists known for personal iconography, Hieronymus Bosch, Odilon Redon, the Pre-Raphaelites, wherever dense symbolism challenges traditional cataloging. 111

I want to conclude here to be clear about what this pilot establishes, and what it does not. This pilot establishes three things: that multimodal embeddings combined with efficient similarity indexing can support sub-millisecond retrieval over a stratified Blake corpus; that performance degrades predictably across object-, motif-, and iconographic-level tagging, with the gradient itself a usable empirical finding for archives weighing adoption; and that the integration of AI tags into an existing scholarly archive can be operationalized concretely through provenance, confidence, version, and audit-log fields. It does not establish that the system interprets Blake, that its outputs generalize beyond the 200-work sample, or that the Tier-3 figures will hold under multi-annotator review. Read against the three-tier framework, success in this study has a specifically digital-humanities meaning: improved discovery without substituting for interpretation; transparent provenance for every machine-generated claim; measurable improvement under human-in-the-loop review; and the avoidance of interpretive foreclosure. AI here is a perceptual assistant under a scholar's supervision, useful precisely because its limits are now mapped, not because they have been overcome. 112

Appendix A: Technical Implementation Details

Note: This appendix contains system architecture, performance benchmarks, and evaluation protocols. 113

A.1 System Architecture

The AI pipeline for analyzing Blake's paintings consists of several integrated components designed for both research experimentation and production deployment. The architecture follows a modular design pattern enabling independent development and testing of each component. 114

Data Ingestion Module: 115

- Interfaces with the William Blake Archive API to retrieve images and metadata
- Handles authentication, rate limiting, and data validation
- Supports batch processing for large-scale corpus analysis

Preprocessing Pipeline: 116

- Image standardization: Resize to 224×224 pixels (CLIP ViT-B/32 requirement)
- Normalization: ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
- Format handling: Support for JPEG, PNG, TIFF formats

- Metadata processing: XML parsing for TEI-encoded descriptions, text normalization

Embedding Generation: [117]

- CLIP model: OpenAI ViT-B/32 architecture
- Output: 512-dimensional normalized vectors
- Batch processing: 32 images per batch for GPU efficiency
- Storage: HDF5 format for efficient large-scale storage and retrieval

Indexing System: [118]

- FAISS implementation: IndexFlatIP for cosine similarity
- Metadata database: SQLite for development, PostgreSQL for production
- Caching layer: Redis for frequently accessed embeddings

A.2 Performance Benchmarks

Metric	Value
Embedding generation	0.15s per image (GPU)
Index construction (200 works)	2.3s
Query latency	0.001s average
Throughput	44,000+ queries/second
Memory footprint	<100MB (200 works)
Projected 715-work memory	~280MB

Table 10. System Performance Metrics

Task	Metric	Tier 1	Tier 2	Tier 3
Tagging	Top-1	78.3%	62.1%	41.2%
Tagging	Top-5	94.2%	85.7%	68.4%
Tagging	mAP	0.72	0.58	0.39
Retrieval	P@10	0.73	—	—
Retrieval	NDCG@10	0.68	—	—

Table 11. Model Accuracy on Blake Test Set (n=40)

A.3 Evaluation Protocols

Tagging Evaluation: [119]

```
For each test image:
   Generate top-K tag predictions
   Compare against ground truth labels
   Compute precision, recall, F1
   Compute mean average precision
```

Retrieval Evaluation:

120

```
For each query image:
   Retrieve top-K similar images
   Compare against ground truth similarity judgments
   Compute Precision@K, Recall@K, NDCG@K
```

A.4 API Response Examples

Similarity Search Response:

121


```
{
   "query_type": "image_similarity",
   "model": "clip-vit-b32",
   "corpus_size": 200,
   "results": [
      {
         "desc_id": "but812.1.wc.01",
         "title": "The Vision of the Last Judgement (1808)",
         "similarity_score": 0.924,
         "metadata": {
            "medium": "Pen and watercolor",
            "repository": "Petworth House",
            "date": "1808"
         }
      }
   ],
   "execution_time_ms": 1.2
}
```


Few-Shot Learning Response:

122


```
{
   "label": "outstretched_arms",
   "strategy": "mean",
   "support_samples": 3,
   "results": [
      {
         "desc_id": "but303.1.pt.01",
         "title": "Albion Rose",
         "confidence": 0.892
      }
   ]
}
```


A.5 Corpus Manifest

The 200-work corpus includes:

123

Illuminated Manuscripts (56 works):

124

- Songs of Innocence and of Experience (selected plates)
- America a Prophecy
- Europe a Prophecy
- The Book of Ahania
- The Book of Los
- Milton a Poem
- Jerusalem
- The Marriage of Heaven and Hell
- Visions of the Daughters of Albion
- The [First] Book of Urizen

Drawings and Paintings (94 works): 125

- The Vision of the Last Judgement (multiple versions)
- Large Color Prints (Newton, Nebuchadnezzar, etc.)
- Biblical watercolors
- Mythological scenes
- Portrait studies

Commercial Illustrations (30 works): 126

- Illustrations of the Book of Job
- Dante's Divine Comedy illustrations
- Night Thoughts (selected)
- Milton's Paradise Lost illustrations

Other (20 works): 127

- Letters with visual content
- Notebook sketches
- Miscellaneous drawings

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